Code:

```
abstract class EnergySource {
    String s_Id;
    String s_Name;
    double e_Gen;
    EnergySource(String s_Id, String s_Name, double e_Gen) {
        this.s_Id = s_Id;
        this.s_Name = s_Name;
        this.e_Gen = e_Gen;
    }
    abstract double calculateEfficiency();
    void displayDetails() {
        System.out.println("Source ID: " + s_Id);
        System.out.println("Source Name: " + s_Name);
        System.out.println("Energy Generated: " + e_Gen + "
kWh");
        System.out.printf("Efficiency: %.2f\n",
calculateEfficiency());
        System.out.println("      ");
    }
}
class SolarEnergy extends EnergySource {
    SolarEnergy(String s_Id, String s_Name, double e_Gen) {
        super(s_Id, s_Name, e_Gen);
    }
    double calculateEfficiency() {
        return (e_Gen / 5000.0) * 100.0;
    }
}
class WindEnergy extends EnergySource {
    WindEnergy(String s_Id, String s_Name, double e_Gen) {
        super(s_Id, s_Name, e_Gen);
    }
    double calculateEfficiency() {
        return (e_Gen / 8000.0) * 100.0;
    }
}
public class EnergyManagementSystem {
    public static void main(String[] args) {
        EnergySource ref;
        ref = new SolarEnergy("SOL-101", "Kshitij Solar
Energy", 4200.0);
        ref.displayDetails();
        ref = new WindEnergy("WND-202", "Anupam Wind Energy
Center", 5200.0);
        ref.displayDetails();
        System.out.println("Coded by: Kshitij Thakur");
```

```
        System.out.println("Section: C");
        System.out.println("USN: 1BF25CS130");
    }
}
```

Output:

```
 ----jGRASP exec: java EnergyManagementSystem
Source ID: SOL-101
Source Name: Kshitij Solar Energy
Energy Generated: 4200.0 kWh
Efficiency: 84.00

Source ID: WND-202
Source Name: Anupam Wind Energy Center
Energy Generated: 5200.0 kWh
Efficiency: 65.00

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Section: C
USN: 1BF25CS130

 ----jGRASP: Operation complete.
```